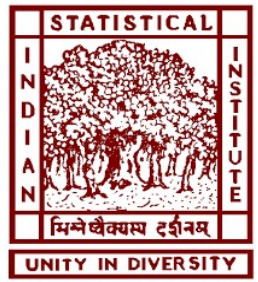
Advanced Statistical Methods II

Lecture 24: Partial Least Squares and Least Angle Regression

Monitirtha Dey

BSDS Third Year Students

Indian Statistical Institute



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Outline of Today's Lecture

Where we are: two questions

Partial least squares

Least angle regression

Summary

Recap : Regression model in PC coordinates

Model: $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{X}$ centred, $\operatorname{rank} \boldsymbol{X} = p$, $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$. Then $\widehat{\beta}_0 = \bar{y}$, and we may take $\boldsymbol{y}$ centred. Define

$$\boldsymbol{\gamma} = \boldsymbol{V}^\top \boldsymbol{\beta} \ \text{(coordinates of } \boldsymbol{\beta} \text{ along the PCs)}, \qquad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{D} \ \text{(scores)}.$$

Since $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{Z}\boldsymbol{\gamma}$, the model is $\boldsymbol{y} = \beta_0 \mathbf{1} + \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, with orthogonal regressors $\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{D}^2$.

OLS in these coordinates:

$$\widehat{\boldsymbol{\gamma}} = (\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{D}^{-1} \boldsymbol{U}^\top \boldsymbol{y}, \qquad \widehat{\gamma}_j = \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j}, \qquad \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\gamma}}) = \sigma^2 \boldsymbol{D}^{-2}.$$

- The $\widehat{\gamma}_j$ are uncorrelated (independent under normality), $\mathbb{E}\widehat{\gamma}_j = \gamma_j$, $\mathbb{V}\mathrm{ar}(\widehat{\gamma}_j) = \sigma^2 / d_j^2$.
- $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{V}\widehat{\boldsymbol{\gamma}} = \sum_{j=1}^{p} \widehat{\gamma}_j \boldsymbol{v}_j$.

Recap: Principal components regression

PCR with M components ($0 \leq M \leq p$): regress $\boldsymbol{y}$ on the first M score vectors $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_M$, and transform back:

$$\widehat{\boldsymbol{\beta}}_M = \sum_{j=1}^{M} \widehat{\gamma}_j \, \boldsymbol{v}_j = \sum_{j=1}^{M} \frac{\boldsymbol{u}_j^\top \boldsymbol{y}}{d_j} \, \boldsymbol{v}_j = \boldsymbol{V}_M \boldsymbol{D}_M^{-1} \boldsymbol{U}_M^\top \boldsymbol{y}.$$

- Because the scores are orthogonal, $\widehat{\gamma}_j$ does not depend on which other components are in the model: PCR fits are *nested*.
- $\widehat{\boldsymbol{\beta}}_p = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$; $\widehat{\boldsymbol{\beta}}_0 = \mathbf{0}$ (intercept-only model).
- $\widehat{\boldsymbol{\beta}}_M$ is a coefficient vector for *all* p original predictors: PCR reduces dimension but does *not* select variables.
- The directions $\boldsymbol{v}_j$ and their order are determined by $\boldsymbol{X}$ alone; $\boldsymbol{y}$ enters only through $\widehat{\gamma}_1, \ldots, \widehat{\gamma}_M$.

Recap

In the population,

$$\mathbb{C}\mathrm{orr}^2(Y,\, Y_j) = \frac{\lambda_j c_j^2}{\mathbb{V}\mathrm{ar}(Y)}, \qquad c_j = \boldsymbol{\gamma}_j^\top \boldsymbol{\beta},$$

and nothing forces $\lambda_j c_j^2$ to be large when λ_j is large.

Here $Y_j = \gamma_j^\mathsf{T}(\mathbf{X} - \mu)$.

Two questions, two methods

Question A: relevance

Can we build a few derived directions $\boldsymbol{X}\boldsymbol{w}_1, \ldots, \boldsymbol{X}\boldsymbol{w}_M$ that are *stable* (high variance, like PCs) *and relevant* for $\boldsymbol{y}$?

Answer: partial least squares (PLS) (Wold, 1966; Wold et al., 1984).

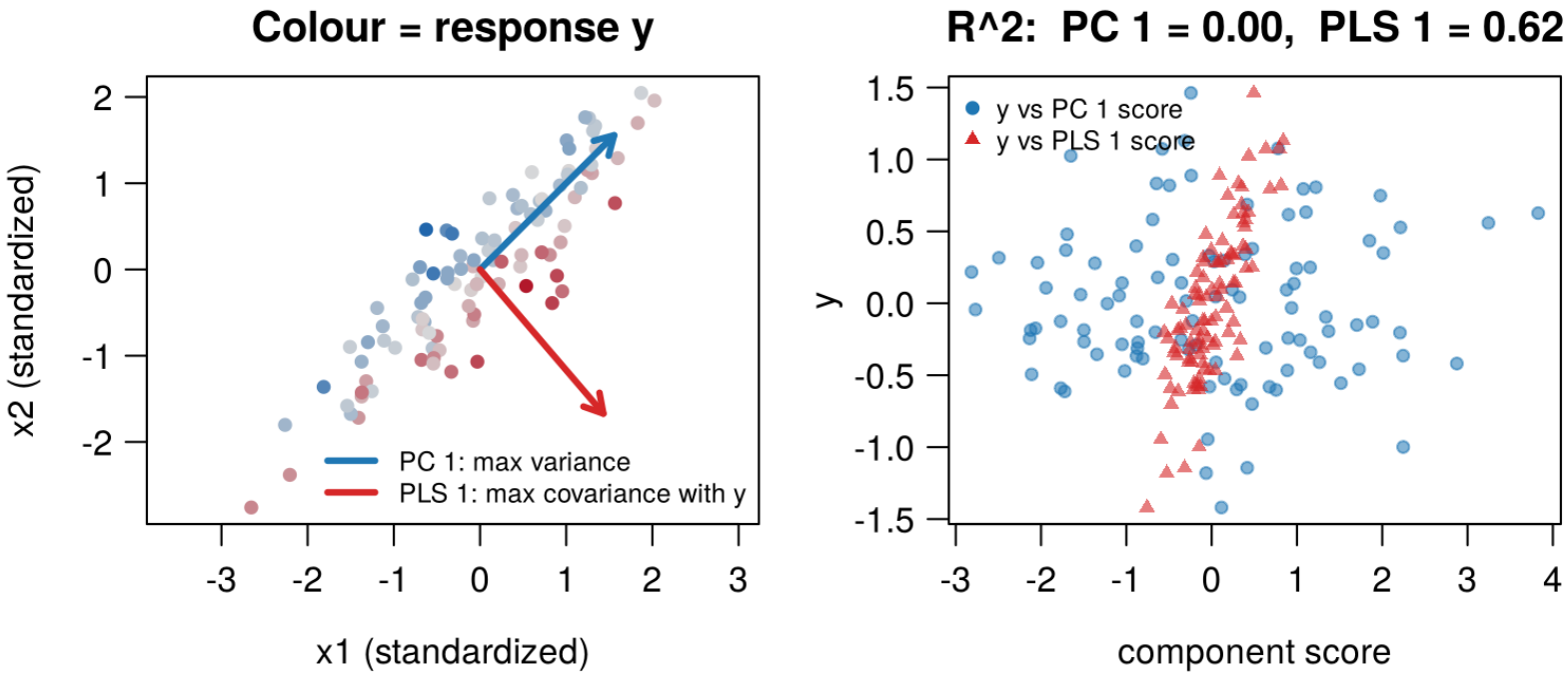
Question B: sparsity

Can we build a model that uses only a *few of the original predictors*, chosen with $\boldsymbol{y}$ in mind, without searching all 2^p subsets and without the greediness of forward selection?

Answer: least angle regression (LARS) (Efron et al., 2004), closely tied to the lasso (Tibshirani, 1996).

Both methods are driven by the same object: the vector of current correlations $\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$ between the predictors and the current residual. PLS moves in the direction of the *whole* vector $\widehat{\boldsymbol{c}}$; LARS moves only along the predictors with the *largest* $|\widehat{c}_j|$.

Motivation: the direction PCR misses



Two standardized predictors with correlation 0.9, $n = 100$, $y = x_1 - x_2 + \text{noise}$. The response varies along the *short* axis of the cloud. PC 1 (variance 1.9) explains none of y. The direction $\boldsymbol{w}_1 \propto \boldsymbol{X}^\top \boldsymbol{y}$ explains 62% of the variance of y. How do we find such directions in general?

Partial Least Squares

Three criteria for one direction

For a unit vector $\boldsymbol{w}$, consider the derived variable $\boldsymbol{t} = \boldsymbol{X}\boldsymbol{w}$. Three natural goals:

Method	maximize over $\|\boldsymbol{w}\| = 1$	what it rewards
OLS	$\mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$	fit only
PCR (its directions = PCs)	$\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$	stability only
PLS	$\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$	both

Indeed

$$\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{y}).$$

- OLS: the best-fitting direction may be one in which $\boldsymbol{X}$ hardly varies (unstable).
- PCR: a stable direction may be irrelevant for $\boldsymbol{y}$.
- PLS: a direction must be *both* correlated with $\boldsymbol{y}$ *and* well supported by the data.
- Stone and Brooks (1990): *continuum regression* maximizes $\mathbb{C}\mathrm{orr}^2 \cdot \mathbb{V}\mathrm{ar}^{\delta}$, $\delta \geq 0$, and moves continuously from OLS ($\delta = 0$) through PLS ($\delta = 1$) to PCR ($\delta \to \infty$).

The first PLS direction

Proposition

$$\boldsymbol{w}_1 = \arg\max_{\|\boldsymbol{w}\|=1} \mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \frac{\boldsymbol{s}}{\|\boldsymbol{s}\|}, \qquad \boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y}.$$

Proof. With centred data, $(n-1)\mathbb{C}\mathrm{ov}(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{w}^\top \boldsymbol{s}$. By Cauchy–Schwarz, $(\boldsymbol{w}^\top \boldsymbol{s})^2 \le \|\boldsymbol{w}\|^2 \|\boldsymbol{s}\|^2 = \|\boldsymbol{s}\|^2$, with equality iff $\boldsymbol{w} = \pm \boldsymbol{s}/\|\boldsymbol{s}\|$. □

- $s_j = \boldsymbol{x}_j^\top \boldsymbol{y}$: the jth coordinate of $\boldsymbol{w}_1$ is the covariance of x_j with y. No matrix is inverted, so it works for any p, including $p > n$.
- **Population version:** $\boldsymbol{w}_1 \propto \mathbb{C}\mathrm{ov}(\boldsymbol{X}, Y) = \boldsymbol{\Sigma}\boldsymbol{\beta}$.

The first PLS direction in PC coordinates

Write $\boldsymbol{s}$ in the eigenbasis of $\boldsymbol{X}^\top \boldsymbol{X}$. Since $\boldsymbol{X}^\top \boldsymbol{u}_j = d_j \boldsymbol{v}_j$ and $\widehat{\gamma}_j = \boldsymbol{u}_j^\top \boldsymbol{y}/d_j$,

$$\boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y} = \sum_{j=1}^{p} d_j (\boldsymbol{u}_j^\top \boldsymbol{y})\, \boldsymbol{v}_j = \sum_{j=1}^{p} d_j^2\, \widehat{\gamma}_j\, \boldsymbol{v}_j.$$

In the population: $\boldsymbol{\Sigma}\boldsymbol{\beta} = \sum_j \lambda_j c_j \boldsymbol{\gamma}_j$.

- PCR weighs PC j by its variance d_j^2 (keep or kill).
- PLS weighs PC j by *variance* $\times$ *estimated coefficient*, $d_j^2 \widehat{\gamma}_j$: a high-variance PC that is irrelevant for $\boldsymbol{y}$ gets a small weight, and a moderately small PC that is relevant still contributes.
- **Uncorrelated standardized predictors** ($\boldsymbol{\Sigma} = \boldsymbol{I}$): then $\boldsymbol{w}_1 \propto \boldsymbol{\Sigma}\boldsymbol{\beta} = \boldsymbol{\beta}$. The first PLS direction points exactly at $\boldsymbol{\beta}$, while the PCs are not even unique (Lecture 23).

One direction is rarely enough. How should the next directions be chosen?

Further PLS directions

PLS directions (Hastie et al., 2009, Sec. 3.5.2). For $m = 2, 3, \ldots,$

$$\boldsymbol{w}_m = \arg\max_{\boldsymbol{w}} \ \mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})$$

subject to $\|\boldsymbol{w}\| = 1$ and $\mathbb{C}\mathrm{ov}(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{X}\boldsymbol{w}_k) = 0$ for $k < m$.
PLS regression with M components: regress $\boldsymbol{y}$ on the scores $\boldsymbol{t}_m = \boldsymbol{X}\boldsymbol{w}_m$, $m \leq M$; equivalently, least squares over $\boldsymbol{\beta} \in \mathrm{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M)$:

$$\widehat{\boldsymbol{\beta}}_{\mathrm{PLS}}(M) = \boldsymbol{W}_M(\boldsymbol{W}_M^\top \boldsymbol{S}\boldsymbol{W}_M)^{-1}\boldsymbol{W}_M^\top \boldsymbol{s},$$

where $\boldsymbol{S} = \boldsymbol{X}^\top\boldsymbol{X}$, $\boldsymbol{s} = \boldsymbol{X}^\top\boldsymbol{y}$ and $\boldsymbol{W}_M = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M]$.

- The scores are mutually uncorrelated, like PC scores, but each one is chosen for its covariance with $\boldsymbol{y}$.
- Compare: PCs maximize $\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$ under the same constraints.
- The formula for $\widehat{\boldsymbol{\beta}}$: put $\boldsymbol{\beta} = \boldsymbol{W}_M\boldsymbol{a}$ and minimize $\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{W}_M\boldsymbol{a}\|^2$ over $\boldsymbol{a}$.

A small example

Centred data, $n = 4$, $p = 2$: $\boldsymbol{x}_1 = (-3, -1, 1, 3)^\top$, $\boldsymbol{x}_2 = (-1, -3, 3, 1)^\top$, $\boldsymbol{y} = (-3, 1, -1, 3)^\top$.

$$\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X} = \begin{pmatrix} 20 & 12 \\ 12 & 20 \end{pmatrix}, \qquad \boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y} = \begin{pmatrix} 16 \\ 0 \end{pmatrix}, \qquad \|\boldsymbol{y}\|^2 = 20.$$

- PCs: $\boldsymbol{v}_1 = (1, 1)^\top / \sqrt{2}$ ($d_1^2 = 32$), $\boldsymbol{v}_2 = (1, -1)^\top / \sqrt{2}$ ($d_2^2 = 8$).
- PLS: $\boldsymbol{w}_1 = \boldsymbol{s}/\|\boldsymbol{s}\| = (1, 0)^\top$, i.e. $\boldsymbol{t}_1 = \boldsymbol{x}_1$. Neither PC 1 nor PC 2.

One component	$\widehat{\beta}$	RSS	R^2
PCR	(0.25, 0.25)	16	0.20
PLS	(0.80, 0.00)	7.2	0.64
OLS (two components)	(1.25, −0.75)	0	1

PC 1 has 80% of the variance of $\boldsymbol{X}$ but explains 20% of $\boldsymbol{y}$.

The structure of PLS: Krylov spaces

Let $\boldsymbol{S} = \boldsymbol{X}^\top \boldsymbol{X}$, $\boldsymbol{s} = \boldsymbol{X}^\top \boldsymbol{y}$, and define the *Krylov spaces*

$$\mathcal{K}_M = \text{span}\{\boldsymbol{s},\ \boldsymbol{S}\boldsymbol{s},\ \boldsymbol{S}^2\boldsymbol{s},\ \ldots,\ \boldsymbol{S}^{M-1}\boldsymbol{s}\}, \qquad \mathcal{K}_1 \subseteq \mathcal{K}_2 \subseteq \cdots.$$

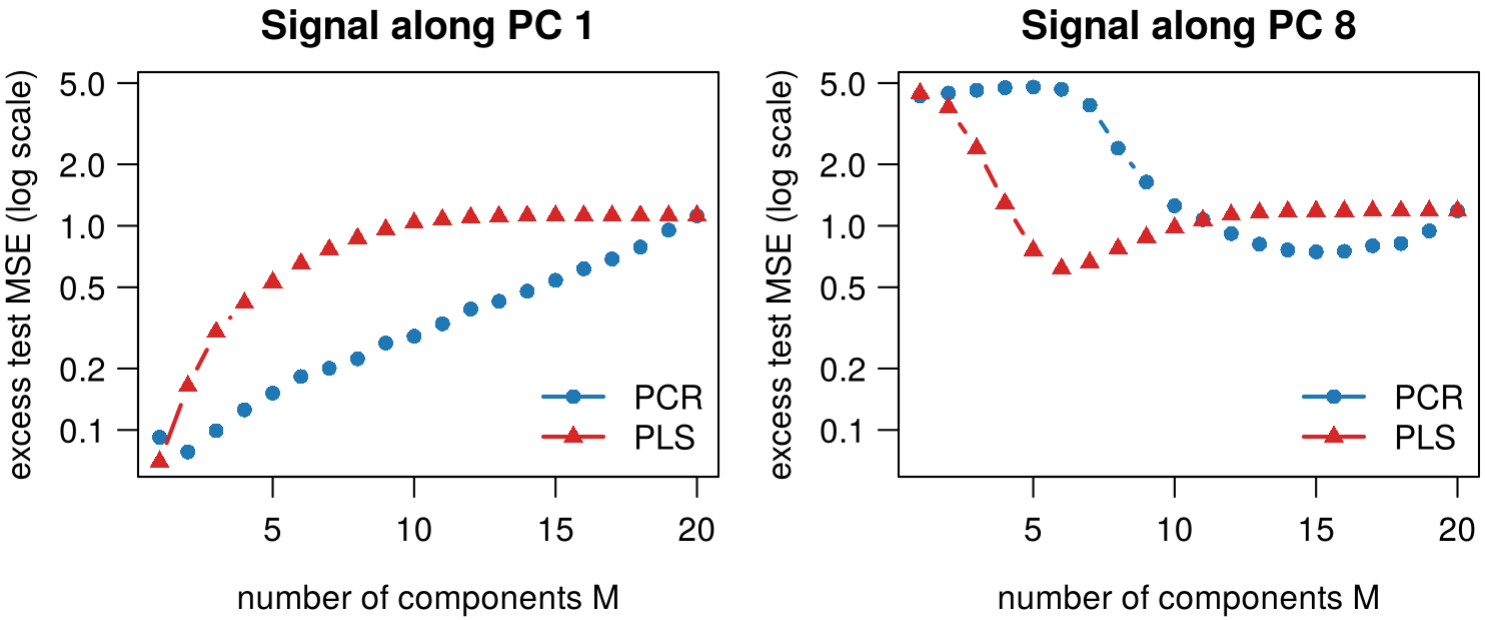
Theorem (Helland, 1988)

$\text{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M) = \mathcal{K}_M$, and therefore

$$\widehat{\boldsymbol{\beta}}_{\text{PLS}}(M) = \underset{\boldsymbol{\beta} \in \mathcal{K}_M}{\arg\min} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2.$$

- PCR: least squares over $\text{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_M)$, a subspace built from $\boldsymbol{S}$ only.
- PLS: least squares over $\mathcal{K}_M$, a subspace built from $\boldsymbol{S}$ *and* $\boldsymbol{s}$.

PCR versus PLS: a simulation



$p = 20$ predictors with AR(1) correlation 0.9, $n = 40$, noise variance 1, 100 replications; β points along population PC 1 (left) or PC 8 (right), with signal variance 4 in both.

- Signal on PC 1: PCR best at $M = 2$ (excess test MSE 0.08), PLS at $M = 1$ (0.07). OLS: 1.12.
- Signal on PC 8: PCR needs $M = 15$ (0.75), PLS $M = 6$ (0.62). OLS: 1.18.

Least Angle Regression

Motivation: sparse models

PCR and PLS use *all* p predictors. Often we want a model with *few* predictors:

- **interpretation**: which baseline measurements matter for disease progression?
- **cost**: each predictor in the final model must be measured for future cases;
- **prediction**: when only a few predictors carry signal, dropping the rest removes their variance.

How to find a good subset?

- **Best subset** for each size k: $\binom{p}{k}$ least-squares fits, 2^p in all; infeasible beyond $p \approx 40$ even with clever search (Furnival and Wilson, 1974).
- **Forward selection**: add one variable at a time, then refit by least squares (Miller, 2002). Cheap, but greedy.
- **Forward stagewise**: take many tiny steps. Cautious, but slow.

LARS is "a less greedy version of forward selection" and "a fast version of forward stagewise" (Efron et al., 2004). To see why, we look at the two in terms of correlations.

Setting and two classical procedures

Standardize: $\sum_i x_{ij} = 0$, $\sum_i x_{ij}^2 = 1$, $\boldsymbol{y}$ centred. For a current fit $\widehat{\boldsymbol{\mu}} = \boldsymbol{X}\widehat{\boldsymbol{\beta}}$ define the current correlations

$$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}), \qquad \widehat{c}_j = \boldsymbol{x}_j^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}) \;\propto\; \mathbb{C}\mathrm{orr}(\boldsymbol{x}_j,\ \text{residual}).$$

Forward selection. Start with $\widehat{\boldsymbol{\mu}} = \boldsymbol{0}$, $\mathcal{A} = \emptyset$. Repeat:

- add $j = \arg\max_{j \notin \mathcal{A}} |\widehat{c}_j|$ to the active set $\mathcal{A}$;
- set $\widehat{\boldsymbol{\mu}} =$ least-squares fit of $\boldsymbol{y}$ on $\{\boldsymbol{x}_j : j \in \mathcal{A}\}$.

Problem: a predictor that enters immediately gets its full least-squares coefficient. A predictor correlated with it then loses most of its correlation with the residual and may never enter, even if it is useful.

Forward stagewise. Start with $\widehat{\boldsymbol{\mu}} = \boldsymbol{0}$. Repeat, with a small $\varepsilon > 0$:

$$j = \arg\max_k |\widehat{c}_k|, \qquad \widehat{\boldsymbol{\mu}} \leftarrow \widehat{\boldsymbol{\mu}} + \varepsilon \,\mathrm{sign}(\widehat{c}_j)\, \boldsymbol{x}_j.$$

Advantage: very cautious, and correlated predictors can share the fit. *Problem:* thousands of steps, and the answer depends on ε.

The LARS idea

Look at what forward stagewise does with small steps:

1. It moves along $\boldsymbol{x}_{j_1}$, the predictor most correlated with $\boldsymbol{y}$. Its correlation with the residual falls.
2. Eventually another predictor $\boldsymbol{x}_{j_2}$ is *equally* correlated with the residual. Stagewise then alternates between the two, in tiny steps.
3. In the limit $\varepsilon \to 0$, it moves along the direction that keeps the two correlations *equal*.

Least angle regression (Efron et al., 2004) does this in a few large steps:

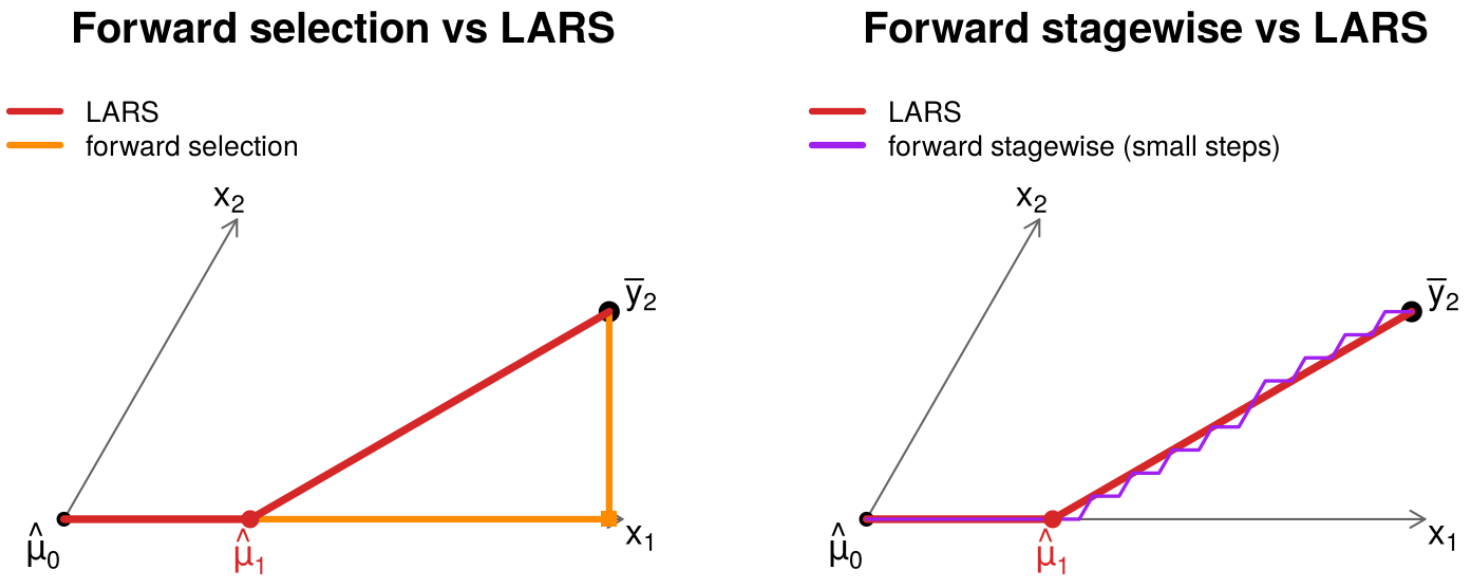
- move along the most correlated predictor *as far as possible*, i.e. until a second predictor has the same absolute correlation with the residual;
- then move along the equiangular direction of the active predictors (equal angles with all of them) until a third predictor catches up; and so on.

After $\min(p, n-1)$ steps all predictors are active; if $p < n$, the last step ends at the OLS fit.

LARS with two predictors

The LARS procedure works roughly as follows. As with classic Forward Selection, we start with all coefficients equal to zero, and find the predictor most correlated with the response, say x_{j_1}. We take the largest step possible in the direction of this predictor until some other predictor, say x_{j_2}, has as much correlation with the current residual. At this point LARS parts company with Forward Selection. Instead of continuing along x_{j_1}, LARS proceeds in a direction equiangular between the two predictors until a third variable x_{j_3} earns its way into the "most correlated" set. LARS then proceeds equiangularly between x_{j_1}, x_{j_2} and x_{j_3}, that is, along the "least angle direction," until a fourth variable enters, and so on.

LARS with two predictors



The plane spanned by unit vectors x_1, x_2; $\bar{y}_2$ is the projection of y onto it. LARS (red) moves along x_1 until the residual makes equal angles with x_1 and x_2 (dotted bisector), then along the bisector to $\bar{y}_2$. **Left:** forward selection jumps to the projection on x_1 first. **Right:** forward stagewise zig-zags around the LARS path in small steps.

Two predictors in formulas

Standardized x_1, x_2 with $x_1^\top x_2 = \rho \in (-1, 1)$, and $c_j = x_j^\top y$ with $c_1 > c_2 > \rho c_1$, $c_2 > 0$.

Step 1. $\widehat{\mu}(\gamma) = \gamma x_1$, so the correlations with the residual are

$$c_1(\gamma) = c_1 - \gamma, \qquad c_2(\gamma) = c_2 - \rho\gamma.$$

They meet at $\widehat{\gamma}_1 = \dfrac{c_1 - c_2}{1 - \rho}$; there x_2 enters, with $\widehat{\beta} = (\widehat{\gamma}_1, 0)$.

Step 2. Move along the bisector $u = (x_1 + x_2)/\sqrt{2(1+\rho)}$ until both correlations vanish, which is the OLS fit

$$\widehat{\beta}_{\mathrm{OLS}} = \frac{1}{1 - \rho^2}\big(c_1 - \rho c_2,\ c_2 - \rho c_1\big).$$

Forward selection instead sets $\widehat{\beta}_1 = c_1$ at step 1. Since

$$c_1 - \widehat{\gamma}_1 = \frac{c_2 - \rho c_1}{1 - \rho} > 0,$$

LARS takes a *shorter* first step: it stops as soon as x_2 is as useful as x_1, and lets the two share the fit from then on.

The equiangular direction

Active set $\mathcal{A}$, signs $s_j = \operatorname{sign}(\widehat{c}_j)$, $\boldsymbol{X}_{\mathcal{A}} = [s_j \boldsymbol{x}_j]_{j \in \mathcal{A}}$, $\boldsymbol{G}_{\mathcal{A}} = \boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{X}_{\mathcal{A}}$, $\mathbf{1}_{\mathcal{A}}$ a vector of ones. Define

$$A_{\mathcal{A}} = \left(\mathbf{1}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}}\right)^{-1/2}, \qquad \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}} \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}}, \qquad \boldsymbol{u}_{\mathcal{A}} = \boldsymbol{X}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}}.$$

Lemma

$\boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{u}_{\mathcal{A}} = A_{\mathcal{A}} \mathbf{1}_{\mathcal{A}}$ and $\|\boldsymbol{u}_{\mathcal{A}}\| = 1$: the unit vector $\boldsymbol{u}_{\mathcal{A}}$ makes the same angle with every (signed) active predictor.

Proof. $\boldsymbol{X}_{\mathcal{A}}^\top \boldsymbol{u}_{\mathcal{A}} = \boldsymbol{G}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}} \mathbf{1}_{\mathcal{A}}$, and $\|\boldsymbol{u}_{\mathcal{A}}\|^2 = \boldsymbol{w}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}} \boldsymbol{w}_{\mathcal{A}} = A_{\mathcal{A}}^2 \, \mathbf{1}_{\mathcal{A}}^\top \boldsymbol{G}_{\mathcal{A}}^{-1} \mathbf{1}_{\mathcal{A}} = 1$. □

The LARS step: $\widehat{\boldsymbol{\mu}} \leftarrow \widehat{\boldsymbol{\mu}} + \widehat{\gamma} \, \boldsymbol{u}_{\mathcal{A}}$, where $\widehat{\gamma} > 0$ is the smallest step at which some inactive predictor reaches the same absolute correlation as the active ones; that predictor then joins $\mathcal{A}$. (An explicit formula for $\widehat{\gamma}$ follows from the next slide; the whole path costs about as much as one least-squares fit.)

The key property: equal, decreasing correlations

Proposition

Let $\widehat{C} = \max_j |\widehat{c}_j|$, attained exactly on $\mathcal{A}$. Along $\widehat{\boldsymbol{\mu}}(\gamma) = \widehat{\boldsymbol{\mu}} + \gamma \boldsymbol{u}_{\mathcal{A}}$,

$$|c_j(\gamma)| = \widehat{C} - \gamma A_{\mathcal{A}} \quad \text{for all } j \in \mathcal{A},$$

so the active predictors keep *equal* absolute correlations with the residual, which *decrease linearly*, and remain larger than those of the inactive predictors until the next one joins.

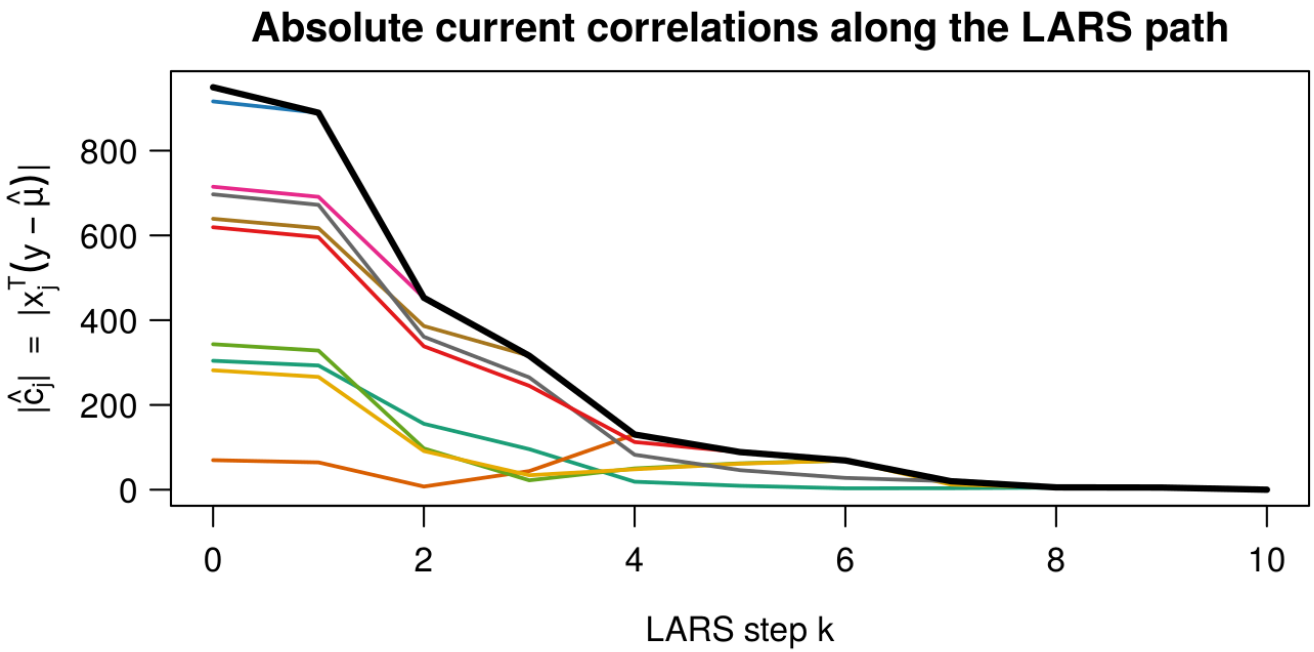
Proof. For $j \in \mathcal{A}$, $c_j(\gamma) = \boldsymbol{x}_j^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}}) - \gamma\, \boldsymbol{x}_j^\top \boldsymbol{u}_{\mathcal{A}} = s_j \widehat{C} - \gamma s_j A_{\mathcal{A}}$, by the lemma. □

- For inactive j: $c_j(\gamma) = \widehat{c}_j - \gamma a_j$, $a_j = \boldsymbol{x}_j^\top \boldsymbol{u}_{\mathcal{A}}$, also linear. Equating $|c_j(\gamma)|$ with $\widehat{C} - \gamma A_{\mathcal{A}}$ gives

$$\widehat{\gamma} = \min_{j \notin \mathcal{A}}{}^{+} \left\{ \frac{\widehat{C} - \widehat{c}_j}{A_{\mathcal{A}} - a_j},\ \frac{\widehat{C} + \widehat{c}_j}{A_{\mathcal{A}} + a_j} \right\} \qquad (\min^{+}: \text{ over positive values}).$$

- The common correlation $\widehat{C}$ decreases from step to step and reaches 0 at the OLS fit.

Example: the diabetes data



$n = 442$ diabetes patients; $p = 10$ baseline variables (age, sex, body mass index, blood pressure, six blood serum measurements); response: a measure of disease progression one year later (Efron et al., 2004). Absolute current correlations $|\widehat{c}_j|$ along the LARS path: the active ones coincide (thick line, the maximum) and decrease linearly; each new variable joins the moment its curve meets the maximum.

Connection to the lasso

Lasso (Tibshirani, 1996):

$$\widehat{\boldsymbol{\beta}}(\lambda) = \arg\min_{\boldsymbol{\beta}} \ \tfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1, \qquad \lambda \geq 0.$$

Optimality (KKT) conditions. $\widehat{\boldsymbol{\beta}}$ is a lasso solution iff

$$\boldsymbol{x}_j^\top(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}) = \lambda \operatorname{sign}(\widehat{\beta}_j) \ \text{ if } \widehat{\beta}_j \neq 0, \qquad |\boldsymbol{x}_j^\top(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}})| \leq \lambda \ \text{ if } \widehat{\beta}_j = 0.$$

- In words: all predictors in the model have the same absolute correlation λ with the residual, and every predictor outside has smaller correlation. This is exactly the LARS property, with $\lambda = \widehat{C}$.
- One extra requirement: the sign of a lasso coefficient must agree with the sign of its correlation (i.e, correlation with the residual), $\operatorname{sign}(\widehat{\beta}_j) = \operatorname{sign}(\widehat{c}_j)$. LARS can violate it: a coefficient may pass through zero.

LARS computes the whole lasso path

Theorem (Efron, Hastie, Johnstone and Tibshirani, 2004)

Modify LARS as follows: if a non-zero coefficient reaches zero during a step, stop there, remove its variable from the active set, and recompute the equiangular direction. Then the modified LARS path is the complete lasso path $\{\widehat{\boldsymbol{\beta}}(\lambda) : \lambda \geq 0\}$ (if variables enter and leave one at a time).

- Another modification (restricting the direction to a cone) gives the limit of forward stagewise as $\varepsilon \to 0$ (Efron et al., 2004; Hastie et al., 2007).
- So LARS, lasso and stagewise are three members of one family. They often give nearly identical paths; they coincide when no coefficient changes sign along the way.
- **Orthonormal predictors** ($\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$): the lasso is soft thresholding,

$$\widehat{\beta}_j(\lambda) = \operatorname{sign}(\widehat{\beta}_j^{\mathrm{OLS}})\big(|\widehat{\beta}_j^{\mathrm{OLS}}| - \lambda\big)_+,$$

and LARS, lasso and stagewise all coincide: the variables enter in the order of $|\boldsymbol{x}_j^\top \boldsymbol{y}|$ and every coefficient grows linearly.

Summary

PLS and LARS side by side

	PLS	LARS
Driving quantity	$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$	$\widehat{\boldsymbol{c}} = \boldsymbol{X}^\top(\boldsymbol{y} - \widehat{\boldsymbol{\mu}})$
Uses of $\widehat{\boldsymbol{c}}$	whole vector: new direction $\propto \widehat{\boldsymbol{c}}$	largest entries only
Model after M steps	all p predictors	M predictors (sparse)
Search space	Krylov space $\mathcal{K}_M$	span of active predictors
Fit within the space	least squares	equiangular, shrunken
End point (for $p < n$)	OLS	OLS
Linear in $\boldsymbol{y}$?	no	no
Related to	PCR	lasso, forward stagewise

- Both build a *sequence* of models of growing complexity, from the null model to OLS, using $\boldsymbol{y}$ at every step; the number of steps is the tuning parameter.
- Both answer the weak spot of PCR: directions (PLS) or variables (LARS) are chosen by their *correlation with the current residual*, not by the variance of $\boldsymbol{X}$.

One design, four methods

Orthonormal predictors, $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}_p$, so $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \boldsymbol{X}^\top \boldsymbol{y}$:

Method	Estimate
PCR	not defined: all $d_j = 1$, so the PCs are not unique
Ridge	$\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}/(1+\lambda)$: uniform shrinkage
PLS	$M = 1$ already gives $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$
LARS / lasso	$\mathrm{sign}(\widehat{\beta}_j^{\mathrm{OLS}})(\lvert\widehat{\beta}_j^{\mathrm{OLS}}\rvert - \lambda)_+$: soft thresholding

- The four methods react to collinearity in different ways; without collinearity, only the lasso still does something (it selects).
- With strong collinearity: PCR drops low-variance directions; ridge shrinks them smoothly; PLS keeps the directions that covary with $\boldsymbol{y}$; LARS/lasso tends to pick one predictor from a group of highly correlated ones.

Take-home messages

- **Motivation for PLS:** PCR picks directions by $\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})$ only. PLS picks them by $\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2 \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w}) \cdot \mathbb{V}\mathrm{ar}(\boldsymbol{y})$: stable *and* relevant.
- First PLS direction $\propto \boldsymbol{X}^\top \boldsymbol{y} = \sum_j d_j^2 \widehat{\gamma}_j \boldsymbol{v}_j$; later directions $\propto$ covariances with the current residual. PLS(M) is least squares over the Krylov space $\mathcal{K}_M$; it reaches OLS after as many steps as $\boldsymbol{S}$ has (relevant) distinct eigenvalues.
- PLS fits better than PCR with the same M, shrinks the norm but can expand single directions, and is non-linear in $\boldsymbol{y}$.
- **Motivation for LARS:** sparse models without 2^p fits; forward selection is too greedy, forward stagewise too slow.
- LARS moves along the equiangular direction of the active set; active correlations stay equal and decrease linearly. With a small modification it gives the entire lasso path; each step costs about one degree of freedom.
- In both, choose the number of steps by C_p or cross-validation.

Practice problems

1. Show that $\mathbb{C}\mathrm{ov}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y}) = \mathbb{C}\mathrm{orr}^2(\boldsymbol{X}\boldsymbol{w}, \boldsymbol{y})\mathbb{V}\mathrm{ar}(\boldsymbol{X}\boldsymbol{w})\mathbb{V}\mathrm{ar}(\boldsymbol{y})$, and that the first PLS direction is $\boldsymbol{s}/\|\boldsymbol{s}\|$.
2. Show that $\boldsymbol{s} = \sum_j d_j^2 \widehat{\gamma}_j \boldsymbol{v}_j$. For $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$, show that PLS(1) = OLS.
3. Prove that PLS(M) equals OLS once M is at least the number of distinct eigenvalues of $\boldsymbol{S}$. Deduce that PLS(2) = OLS for equicorrelated standardized predictors.
4. For $M = 1$, show $f_j = d_j^2/\theta_1$ with $\theta_1 = \boldsymbol{s}^\top \boldsymbol{S}\boldsymbol{s}/\boldsymbol{s}^\top \boldsymbol{s}$, and deduce $f_1 \geq 1 \geq f_p$.
5. Toy data with $\boldsymbol{y} = (-2, 2, -2, 2)^\top$: compute PCR and PLS with one component. Which PC does PLS pick?
6. Prove the equiangular lemma and the equal-correlation proposition for LARS. Derive the formula for $\widehat{\gamma}$.
7. Redo the two-predictor LARS calculation when $0 < c_2 < \rho c_1$. With which sign does $\boldsymbol{x}_2$ enter, and after how long a first step?
8. Derive the lasso KKT conditions. For $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}$, show that the lasso is soft thresholding and that LARS gives the same path.
9. Explain why forward selection after k steps has more than k degrees of freedom.

References

Butler, N. A. and Denham, M. C. (2000). The peculiar shrinkage properties of partial least squares regression. *Journal of the Royal Statistical Society, Series B*, 62(3), 585–593.

de Jong, S. (1993a). SIMPLS: an alternative approach to partial least squares regression. *Chemometrics and Intelligent Laboratory Systems*, 18(3), 251–263.

de Jong, S. (1993b). PLS fits closer than PCR. *Journal of Chemometrics*, 7(6), 551–557.

de Jong, S. (1995). PLS shrinks. *Journal of Chemometrics*, 9(4), 323–326.

Efron, B., Hastie, T., Johnstone, I., and Tibshirani, R. (2004). Least angle regression (with discussion). *The Annals of Statistics*, 32(2), 407–499.

Frank, I. E. and Friedman, J. H. (1993). A statistical view of some chemometrics regression tools. *Technometrics*, 35(2), 109–135.

Furnival, G. M. and Wilson, R. W. (1974). Regressions by leaps and bounds. *Technometrics*, 16(4), 499–511.

Garthwaite, P. H. (1994). An interpretation of partial least squares. *Journal of the American Statistical Association*, 89(425), 122–127.

Hastie, T., Taylor, J., Tibshirani, R., and Walther, G. (2007). Forward stagewise regression and the monotone lasso. *Electronic Journal of Statistics*, 1, 1–29.

Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer, New York.

References (cont.)

Hastie, T., Tibshirani, R., and Wainwright, M. (2015). *Statistical Learning with Sparsity: The Lasso and Generalizations*. CRC Press, Boca Raton, FL.

Helland, I. S. (1988). On the structure of partial least squares regression. *Communications in Statistics – Simulation and Computation*, 17(2), 581–607.

Hestenes, M. R. and Stiefel, E. (1952). Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6), 409–436.

Kalivas, J. H. (1997). Two data sets of near infrared spectra. *Chemometrics and Intelligent Laboratory Systems*, 37(2), 255–259.

Krämer, N. and Sugiyama, M. (2011). The degrees of freedom of partial least squares regression. *Journal of the American Statistical Association*, 106(494), 697–705.

Lingjærde, O. C. and Christophersen, N. (2000). Shrinkage structure of partial least squares. *Scandinavian Journal of Statistics*, 27(3), 459–473.

Mallows, C. L. (1973). Some comments on C_p. *Technometrics*, 15(4), 661–675.

Miller, A. J. (2002). *Subset Selection in Regression*, 2nd ed. Chapman & Hall/CRC, Boca Raton, FL.

Osborne, M. R., Presnell, B., and Turlach, B. A. (2000). On the LASSO and its dual. *Journal of Computational and Graphical Statistics*, 9(2), 319–337.

Phatak, A. and de Hoog, F. (2002). Exploiting the connection between PLS, Lanczos methods and conjugate gradients: alternative proofs of some properties of PLS. *Journal of Chemometrics*, 16(7), 361–367.

References (cont.)

Stone, M. and Brooks, R. J. (1990). Continuum regression: cross-validated sequentially constructed prediction embracing ordinary least squares, partial least squares and principal components regression. *Journal of the Royal Statistical Society, Series B*, 52(2), 237–269.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society, Series B*, 58(1), 267–288.

Wold, H. (1966). Estimation of principal components and related models by iterative least squares. In P. R. Krishnaiah (ed.), *Multivariate Analysis*, pp. 391–420. Academic Press, New York.

Wold, S., Ruhe, A., Wold, H., and Dunn, W. J. (1984). The collinearity problem in linear regression. The partial least squares (PLS) approach to generalized inverses. *SIAM Journal on Scientific and Statistical Computing*, 5(3), 735–743.

Zou, H., Hastie, T., and Tibshirani, R. (2007). On the "degrees of freedom" of the lasso. *The Annals of Statistics*, 35(5), 2173–2192.